Predicting shipment ETA and how delay propagates

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1. What is predicted

The forecast is made when port-of-discharge dwell has just finished. Origin, loading, ocean transit, and discharge have been observed. Customs and inland delivery are still ahead. The brief does not fix this moment. I chose it because upstream delay is then known, and two stages remain in which that delay can be absorbed or passed on.

All clocks are UTC. Final delay, in hours, is the actual end of inland delivery minus the scheduled end. Positive means late. Zero or negative means on time, including early. There is no separate yes-or-no “delayed” label. The predicted arrival is the scheduled arrival plus the predicted delay. Reporting delay in hours satisfies the brief’s request for a likelihood *or* a magnitude.

2. Data, mechanism, and ground truth

The set is synthetic: 200 shipments, seed 20261111, four routes (CNSHA–NLRTM, SGSIN–USLAX, INNSA–AEJEA, KRPUS–DEHAM). Every journey has six stages: origin departure, port of loading, ocean transit, port of discharge, customs, and inland delivery.

The schedule is written first. Each stage has a planned duration plus a fixed buffer, which is the slack. The realized duration is planned time, plus a congestion effect, plus Gaussian noise, floored at one hour. Congestion is drawn uniformly from 0 to 1 on loading, discharge, and customs only. A full congestion index adds 20, 40, and 16 hours on those three stages. Weather is not simulated.

A stage cannot start before its scheduled time, so finishing early does not pull the next milestone forward. Delay carries on only when it is larger than the buffer still left. Ground truth is the actual inland arrival this rule produces. In this seed, 27 of 200 customs milestones are left unrecorded. The clearance still happens; the hidden timestamps are kept only so the mechanism can be audited, and they are not model inputs.

One closed example checks the arithmetic. EX-001, route INNSA–AEJEA, departs 2026-01-15 00:00 UTC with zero noise, discharge congestion 1, and customs congestion 0.5. Discharge finishes 30 hours behind its scheduled end. Customs, starting 30 hours late and running 8 hours long against a 6-hour buffer, passes on 32 hours. Inland slack of 6 hours leaves a final delay of 26 hours: scheduled arrival 2026-01-26 16:00 UTC, actual arrival 2026-01-27 18:00 UTC. A unit test fails if this changes.

Across the 200 shipments, 48 are on time and 152 are late. Final delay runs from −11.9 to 49.1 hours (median 7.2, mean 10.0). Forty-two of the on-time shipments were late at an earlier stage, and a later buffer absorbed it. That is the behaviour the model is meant to respect.

3. Method

At the decision time the model may use observed durations through discharge, the published schedule, and the congestion indexes, including customs congestion, which I treat as already known. It may not use any customs or inland truth timestamp.

Each remaining stage has its own ridge regression. The features are planned duration and the congestion index. The buffer is not a regression feature; the propagation rule applies it afterwards. With about 160 training rows per stage and standardized inputs, a penalty of 1 barely moves an ordinary least-squares fit. Ridge is used as a small linear model, not as a heavily shrunk one.

Predicted customs and inland durations are then walked through the same slack rule as the generator. The explanation is an account, not a second model. After the last stage that puts the shipment back on schedule, each later stage contributes its duration minus its scheduled allowance. Stages before that reset contribute zero, because their effect did not survive. Those contributions sum to the predicted delay.

Three comparisons are scored the same way: trust the schedule (predict zero delay), predict the training-set mean delay, and a flat ridge on final delay that sees lateness at discharge, customs congestion, and route, but has no slack rule. The split is five folds by shipment, seed 20261111, so stages of one journey never sit on both sides. Each shipment is scored once. The metric is mean absolute error of final delay in hours. Bias is the mean of prediction minus actual; positive bias means the call is too late.

4. Results

Method	MAE (hours)	Bias (hours)	Fold MAE
Trust the schedule	12.00	−9.97	9.85 to 13.61
Training-set mean delay	10.45	0.00	10.28 to 10.71
Flat ridge on final delay	4.76	−0.05	4.30 to 5.24
Stage durations, then propagation	4.04	−0.07	3.30 to 4.62

Out-of-fold duration error is 2.94 hours for customs and 2.97 hours for inland delivery, about the size of the noise that was added. By route, propagation MAE is 4.54, 3.41, 3.56, and 4.68 hours (CNSHA–NLRTM, INNSA–AEJEA, KRPUS–DEHAM, SGSIN–USLAX), against schedule MAEs of 10.26 to 14.12. Bias near zero means the propagation model is not systematically early or late.

The flat ridge is close. Most of the gain over the timetable comes from already knowing the delay at discharge. I would not justify the structured model by the 0.7-hour MAE gap. It is justified because it names the hours.

SHP-004, INNSA–AEJEA, is the fold-0 holdout whose absolute error is closest to that fold's median, so it is a typical miss rather than a best case. Leaving discharge it is 10.9 hours late. Ocean transit had already put it back on schedule, so origin, loading, and ocean contribute 0. Predicted customs adds 2.7 hours. Predicted inland delivery absorbs 6.7 hours. The sum is 6.9 hours, which is the predicted delay. The actual delay is 10.4 hours, so this call is 3.4 hours too short. Scheduled arrival is 2026-02-26 19:00 UTC, the prediction is 2026-02-27 01:56 UTC, and the actual arrival is 2026-02-27 05:23 UTC.

5. Assumptions and limitations

These limits are part of the result, not a footnote.

1. **The low error agrees with the simulator. It does not describe real ports.** Durations in the data are linear in planned time and congestion, plus noise, and the regression uses those same inputs. A different held-out mechanism, such as ocean weather or random buffers, was never tested. That is the main gap.
2. **The decision time was chosen, not given.** These numbers are not a booking-time ETA. At booking, no stage delay has been observed, and the error would be larger.
3. **Customs congestion is treated as known** when discharge has just ended. If that index is only a forecast, or is missing, customs duration error grows and the final-delay error grows with it.
4. **The explanation is an identity under rules I wrote.** "Discharge contributed 10.9 hours" means that, under the no-early-start rule and the fixed buffers, 10.9 hours of that stage's excess remain in the predicted delay. It is not a causal claim about a real terminal. The buffers are the same on every route, and a stage is not allowed to start early. Both are assumptions.
5. **Missing customs data does not stress this model.** At this decision time customs has not happened, so a hidden milestone is never an input. Incomplete history would matter if the score were made after a missing clearance. I did not build an imputation step. Training labels still use the true customs duration when the milestone was not logged.
6. **The sample is small and the world is narrow.** Two hundred shipments and four invented lanes leave the fold MAE moving between 3.3 and 4.6 hours. There is no weather, strike, blank sailing, rollover, revised schedule, transshipment, or port network.
7. **Accuracy alone would not pick this model over the flat ridge** (4.04 versus 4.76 hours). The flat model cannot say which stage's hours reached the consignee. The propagation model can, and the hours add up to its prediction. That is the reason to keep it.

Regenerate the files from the repository root with `python -m eta_delay` and `python -m eta_delay.evaluate`. The same seed rewrites the same rows. `pytest` checks the 26-hour example, blocks future truth timestamps from entering the features, and checks that the stage hours sum to the predicted delay.